

ESTEVAN, SK, Canada

WMO: 718620

Lat: 49.2050N

Lon: 102.9670W

Elev: 580

StdP: 94.55

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.1	-29.3	-36.7	0.1	-32.0	-33.9	0.2	-29.2	14.4	-11.1	13.2	-12.9	4.3	300	0.710

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	31.1	19.8	29.3	18.9	27.5	18.2	21.6	28.0	20.4	26.9	19.3	25.6	5.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.4	15.2	25.2	18.1	14.0	23.8	17.0	13.0	22.6	65.9	28.0	61.5	26.9	57.5	25.6	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.5	10.8	9.6	DB	-36.1	34.6	3.4	2.1	-38.5	36.0	-40.5	37.2	-42.5	38.4	-45.0	39.9	(4)
(5)			WB	-36.2	23.9	3.4	1.3	-38.6	24.9	-40.6	25.6	-42.5	26.4	-45.0	27.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-13.8	-12.1	-4.8	4.4	11.2	16.4	19.4	18.5	13.2	5.1	-3.8	-11.2	(6)	
	DBStd	13.30	8.57	8.08	7.95	5.85	4.58	3.55	3.11	3.46	4.60	5.22	6.82	7.83	(7)	
	HDD10.0	3292	738	618	458	179	41	2	0	0	20	166	414	656	(8)	
	HDD18.3	5501	997	851	716	417	226	77	25	40	164	409	664	914	(9)	
	CDD10.0	969	0	0	1	13	77	194	291	265	115	14	0	0	(10)	
	CDD18.3	136	0	0	0	0	3	19	58	47	9	0	0	0	(11)	
	CDH23.3	1836	0	0	0	9	87	245	634	649	201	10	0	0	(12)	
	CDH26.7	554	0	0	0	1	19	59	189	226	60	1	0	0	(13)	
(14) Wind	WSAvg	4.7	5.1	4.9	5.0	5.2	5.2	4.5	3.9	3.8	4.2	4.7	4.8	4.9	(14)	
(15) Precipitation	PrecAvg	434	20	15	21	34	57	70	62	55	44	23	15	18	(15)	
	PrecMax	706	56	43	49	110	170	162	170	149	117	82	30	38	(16)	
	PrecMin	235	3	2	0	7	2	11	9	2	7	0	1	3	(17)	
	PrecStd	112	12	9	14	25	39	38	46	38	28	20	8	9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.6	6.0	16.0	24.6	29.3	31.2	32.8	34.2	31.3	24.8	15.3	6.2	(19)	
		MCWB	2.1	3.2	8.2	12.9	15.7	19.9	21.4	20.1	18.4	13.6	8.4	2.7	(20)	
	2%	DB	2.3	2.9	11.5	20.2	25.8	28.3	30.6	31.4	28.4	20.0	10.8	3.4	(21)	
		MCWB	0.6	1.0	6.2	10.7	14.5	18.5	20.7	19.4	16.8	11.7	5.9	1.2	(22)	
	5%	DB	0.6	1.0	7.5	17.5	23.0	26.1	28.8	29.2	25.6	16.6	7.6	1.4	(23)	
		MCWB	-0.6	-0.3	4.0	9.2	12.9	17.2	20.0	18.8	15.7	9.8	3.9	-0.2	(24)	
	10%	DB	-1.8	-0.8	4.4	14.4	20.5	24.1	26.9	27.1	22.4	13.5	4.7	-0.6	(25)	
		MCWB	-2.8	-1.7	2.2	7.7	11.8	16.2	19.1	18.1	14.1	8.3	2.1	-1.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.3	3.3	9.3	13.3	17.5	22.5	23.6	22.3	19.7	14.8	9.0	3.0	(27)	
		MCDB	4.2	5.5	15.6	22.6	25.6	29.1	29.2	29.5	27.4	22.1	14.6	5.9	(28)	
	2%	WB	0.7	1.2	6.5	11.3	15.3	19.9	22.1	20.8	18.1	12.4	6.2	1.3	(29)	
		MCDB	2.1	2.7	11.2	19.0	22.9	25.5	28.2	28.3	25.8	18.9	10.1	3.2	(30)	
	5%	WB	-0.7	-0.2	4.2	9.7	14.0	18.4	20.9	19.7	16.5	10.6	4.2	-0.2	(31)	
		MCDB	0.6	0.9	7.2	16.4	20.8	24.1	26.9	26.9	23.8	15.4	7.3	1.3	(32)	
	10%	WB	-2.7	-1.8	2.3	8.1	12.7	17.1	19.9	18.6	14.9	8.8	2.3	-1.8	(33)	
		MCDB	-1.8	-0.8	4.3	13.7	18.9	22.4	25.8	25.4	21.2	13.1	4.5	-0.7	(34)	
(35) Mean Daily Temperature Range	MDBR		10.1	10.2	10.0	12.2	13.1	11.5	12.8	14.1	13.9	11.8	9.9	9.5	(35)	
	5% DB	MCDBR	11.0	10.3	13.7	17.8	17.8	15.3	15.4	17.6	19.1	17.9	14.2	11.3	(36)	
		MCWBR	9.6	8.7	9.1	9.7	8.6	7.4	7.4	7.8	9.2	10.3	9.7	9.4	(37)	
	5% WB	MCDBR	10.8	10.0	12.8	16.8	14.9	13.0	13.6	15.3	16.6	16.3	13.5	11.1	(38)	
		MCWBR	9.5	8.6	8.8	9.5	7.8	7.1	7.3	7.5	8.5	9.7	9.4	9.3	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.256	0.253	0.273	0.326	0.341	0.356	0.364	0.377	0.332	0.288	0.256	0.240	(40)	
	taud		2.348	2.398	2.497	2.404	2.412	2.409	2.408	2.350	2.445	2.547	2.500	2.422	(41)	
	Ebn at Noon		816	917	954	923	919	903	889	859	870	866	825	799	(42)	
	Edh at Noon		66	80	87	107	111	112	111	112	92	69	57	53	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.48	2.58	3.88	4.96	5.84	6.18	6.68	5.36	4.01	2.42	1.43	1.06	(44)	
	RadStd		0.14	0.25	0.33	0.32	0.41	0.42	0.34	0.38	0.36	0.26	0.20	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page